

Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Knowledge Distillation, and Few-Shot Meta-Adaptation on UCF101

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Abstract—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline that illustrates how one can construct and train a powerful video action recogniser, extend its knowledge over time to new classes without forgetting old ones, adapting to new scenarios, compress it into a lighter model mitigating performance degradation, and learn new classes under a few-shot regime. The proposed pipeline is composed of five linked studies plus a reference study. Each module answers one of the following questions, comparing several methodologies aiming to find the most suitable one for the current framework. (i) *How do we avoid data leakage?* UCF101’s native train-test split leaks 80% of its data across train and test, meaning that different frames from the same video clip are present in both sets. This setting pollutes training, as train and test sets must be completely disjoint. We replace this mechanism with a new group-disjoint split that avoids leakage completely. (ii) *Which backbone should we select?* To select the most suitable backbone (i.e., feature extractor) for the pipeline, we compare four ImageNet-pretrained CNNs (ResNet18, ResNet50, DenseNet121, and VGG19-BN) and a vision transformer (ViT-B/16), all of which are fine-tuned on the current dataset, against a light, custom CNN trained from scratch. All architectures are attached a LSTM module to capture temporal dependencies. Through this comparison, we select a ResNet50+LSTM model (93.9% on the base classes). On the native split the same backbones score 4.3 to 36.7 points higher, and the vision transformer reaches 99.4%, showing how much leakage inflates the results. (iii) *How do we prevent forgetting?* To prevent forgetting on a $10 \rightarrow 15 \rightarrow 20$ class stream, we tested several continual learning strategies, including rehearsal, EWC, and LwF. The best-performing method, rehearsal, obtains an overall accuracy of 89.8%, while naive fine-tuning on new classes collapses to 26.0%. This demonstrates the effectiveness of continual learning approaches when extending knowledge of a model. (iv) *How do we shrink the model?* To transfer knowledge from the large ResNet50+LSTM teacher to a smaller MobileNetV3-Small+LSTM student, we compare three distillation objectives: logits alignment, features alignment (both through cosine similarity and MSE), and a new class-prototype alignment objective, which extracts and aligns class prototypes instead of the whole feature maps. Under full-data regime, cosine feature distillation allows the MobileNetV3-Small student to achieve an accuracy of 82.7%, at $14\times$ fewer parameters than the teacher model; with only 32 training clips per class, the new class-prototype alignment objective raises the student’s accuracy from 61.5% to 67.9%. (v) *How do we learn new classes from*

a few clips? To teach the distilled student five new sports from only eight clips each, we compare three starting points (the plain student, a student fine-tuned on 40 other sports, and a student meta-trained with MAML on the same sports), each adapted with no protection, with replay, or with replay and LwF. Scored with the harmonic mean of new- and old-class accuracy, MAML combined with replay is the best-balanced strategy (0.600), ahead of the fine-tuned control with replay (0.530), which saw the same extra data. (vi) *How can we exploit frozen foundation models?* Vision-language models such as CLIP have been shown to be extremely effective under few-shot regimes. We compare our ResNet50+LSTM against a frozen CLIP encoder with a small MLP adapter. We evaluate the comparison under 1, 2, 4, 8, and 16 samples per class during training. During 1-shot learning, the CLIP+MLP architecture easily achieves an accuracy of 75.0%, against 17.9% for our ResNet50+LSTM, showing the power of these foundation models. To test the generalisation abilities of our approach, a final capstone combines the best strategy of each stage into a single pipeline and evaluates it on real YouTube videos of five unseen sports, using 10 clips per class for training. Under small update budgets the meta-trained student adapts faster than the plain distilled student: after 13 updates it reaches a new-class accuracy of 0.16 on the unseen test videos, while the plain student reaches 0.00, and after 26 updates it reaches 0.36, compared with 0.23 for the plain student and 0.37 for the $14\times$ larger teacher. With 65 updates the plain student catches up and surpasses it (0.57 against 0.47), confirming that meta-learning is most useful when adaptation must be fast. The whole pipeline runs on a single free Colab T4 GPU.

Index Terms—action-recognition, continual learning, catastrophic forgetting, knowledge distillation, meta-learning, few-shot learning, model compression.

I. INTRODUCTION

Deep video models are prone to forget old classes when fine-tuned on a set of new ones, as models’ parameters for old knowledge get overwritten. This phenomenon is known as catastrophic forgetting [1]. Continual training (or life-long training) aims to mitigate this issue, extending current knowledge to new classes while preserving what was already learned [5]. Videos make this remarkably harder than images: a clip is a large spatio-temporal tensor, therefore storing and replaying past data is expensive, and the temporal model on

top of the per-frame spatial feature extractor adds a second layer subjected to representation drift.

A realistic deployment demands several capabilities at once. A model must (a) be evaluated on data it has never seen, (b) begin with strong features, (c) retain existing classes as new ones arrive, (d) remain compact enough to run on modest hardware, and (e) be able to learn a new class from only a handful of examples. Although these capabilities are typically studied in isolation, we examine them *together* within a single pipeline and on a common dataset. The models are trained *directly on real video frames*, so each reported result reflects the model that would ultimately be deployed.

Our pipeline (Fig. 1) is built on UCF101 [14] and it is composed of five linked studies (S1–S5), each of them answering one of the proposed questions by comparing different strategies. We include a sixth study (S6), which compares meta-learning with a different route to few-shot learning: a frozen CLIP foundation model used as backbone.

Our base method follows a *moving freeze boundary* strategy: it is composed of a CNN backbone used to extract spatial features from each frame, of a LSTM used to extract temporal dependencies, and of a classifier head that *grows* as classes arrive.

Main Contributions.

- **An fair measurement protocol.** We quantify the 80% source-group leakage in UCF101’s native train-test split and replace it with a group-disjoint split that avoids leakage.
- **A comprehensive study for each stage:** six backbones (including a vision transformer) to select the best feature extractor, five continual-learning strategies, four distillation objectives, and nine meta-adaptation arms.
- **A fine-tune control that isolates meta-learning:** we compare the MAML starting point with an ordinary fine-tuned starting point trained on the *same* extra sports, so that the gain attributed to meta-learning is not simply due to the additional data.
- **A foundation model reference:** we include in our study a few-shot learning strategy leveraging a frozen CLIP encoder against our ResNet50+LSTM trained on the same few clips.
- **A real-video capstone:** the best strategy selected at each stage is inserted into a single pipeline, which is evaluated over five unseen sports from a few YouTube clips.

II. RELATED WORK

Continual learning (CL). Traditional CL approaches mainly fall into three families. *Regularisation methods* add a penalty to the standard loss function that discourages changing parameters recognized to be important to the old tasks: EWC [1] weights the penalty by the Fisher information, while synaptic-intelligence variants use online importance estimates [2]. *Functional methods* try to keep the model’s outputs stable: for example, Learning without Forgetting (LwF) [3] distills the old model’s own predictions on new data. *Rehearsal methods* store a portion of old samples and use them during

training, along with new samples for new tasks. One can also combine different strategies: iCaRL [4] combines rehearsal with distillation. **Knowledge distillation (KD).** Traditional KD approaches try to inject into a compact model (student) the knowledge of larger one (teacher) by matching their softened logits [6] or their intermediate features [7]. **Meta and few-shot learning.** Few-shot learning aims to recognise new classes from only a handful of labelled examples. Metric-based methods such as prototypical networks [13] classify a sample by its distance to class prototypes computed in an embedding space. Optimisation-based methods such as MAML [8] instead learn an initialisation from which a few gradient steps adapt the model to a new task; first-order MAML drops the second-order term for efficiency. Few-shot class-incremental learning (FSCIL) [9] adds new classes from few examples while retaining a large base set, and reports the harmonic mean of base and new accuracy so that neither side can be ignored; we adopt that metric. Finally, large vision-language models such as CLIP [12] provide general-purpose features that can be adapted to new classes with little or no training; we use one as a reference in Stage 6.

III. SYSTEM, NOTATION, AND PROTOCOL

A. The shared architecture

A clip is composed of $T=16$ RGB frames at 224×224 resolution, ImageNet-normalised. Once the best recognizer is established, it is used for every stage of the study (Fig. 2). This model is composed of a CNN-backbone for spatial feature extraction, a LSTM module for temporal dependencies learning, and a final classification head. The CNN-backbone ϕ is applied to each frame independently, producing a sequence of spatial features $\phi(x_1), \dots, \phi(x_T)$. This sequence is read by the LSTM module, which outputs its final hidden state $h_T \in \mathbb{R}^{256}$, representing the final clip feature map. A final linear head W maps h_T to class scores.

$$h_T = \text{LSTM}(\phi(x_1), \dots, \phi(x_T)), \quad z = W h_T. \quad (1)$$

The head *grows* as the model is expanded to new classes. Adding k classes appends k rows to W and keeps the existing rows.

B. The moving freeze boundary

At each stage a different portion of the model is trained. Stage 2 (best model selection) keeps the early layers of each pre-trained backbone frozen and fine-tunes only its last block, together with the LSTM and the classifier head. Stage 3 (continual learning) trains the whole teacher on the new classes, with the classifier head expanded, while adopting strategies to avoid catastrophic forgetting. Stage 4 (knowledge distillation) freezes the teacher model completely and trains the whole student to match teacher’s performance. Stage 5 (meta learning) freezes the lower student backbone and meta-trains only its last four blocks, the LSTM, and the head, so the old features are protected while the upper layers reshape. Within the same stage every tested methodology trains the *same* portion of the model (through different strategies).

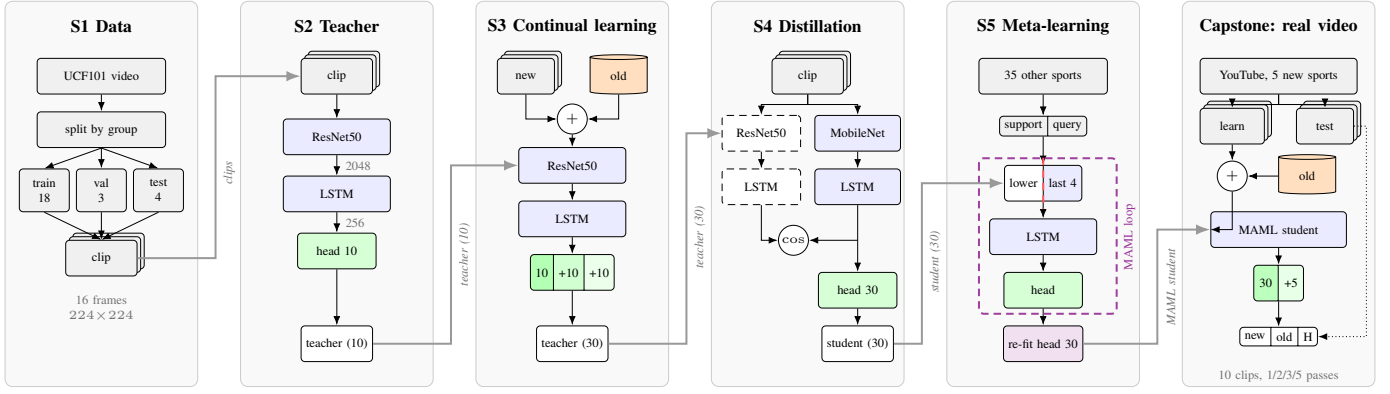


Fig. 1. **The ULAL architecture.** One column per stage, read top to bottom; the grey arrows hand the result of each stage to the part of the next stage it feeds (clips to the teacher, the 10-class teacher to continual learning, the 30-class teacher to distillation, the student to meta-learning, the MAML student to the real-video test). Every column uses the winner of its study (Sections IV–VIII); Section X walks through the whole pipeline. Grey: data; blue: trained; dashed: frozen; orange: stored old clips; violet: meta-learning; green: classifier head; red dashed: freeze boundary.

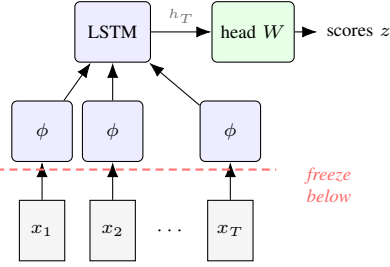


Fig. 2. **The shared recogniser.** The backbone ϕ runs on each of the 16 frames; an LSTM summarises them into a 256-d clip feature h_T ; a growing head classifies. The freeze boundary (dashed) marks how deep training may reach in a stage.

C. Datasets and budget

All experiments use UCF101. Samples are divided into train/val/test with a proportion of $\approx 72/12/16\%$, avoiding data leakage (i.e., frames from the very same clip can’t appear both in train and val or train and test sets). Table I lists the budget of each stage. To ensure reproducibility, seed is fixed before every training run.

D. Evaluation metrics

We refer to *test accuracy* as the fraction of correctly classified clips on the unseen, group-disjoint test set. In the continual-learning and few-shot stages we further split it by class group: *Old* denotes the classes learned in earlier tasks, *New* the classes of the most recent task, and *Base* the original ten classes, which measures long-term retention. When old classes outnumber new ones, a plain average is dominated by the old classes and can reward a model that learns nothing new. For few-shot learning we therefore also report the harmonic mean

$$H = \frac{2 \text{acc}_{\text{new}} \text{acc}_{\text{old}}}{\text{acc}_{\text{new}} + \text{acc}_{\text{old}}}, \quad (2)$$

which is high only when *both* sides are high [9]. Finally, to compare models of different size, we report the number of

TABLE I
BUDGET PER STAGE (SHARED BY EVERY METHOD COMPARED IN THAT STAGE).

Stage	Key settings
Backbone (S2)	Adam, lr 10^{-4} , 5 epochs; last block trained; ViT-B/16: last 2 of 12 blocks, 2-layer LSTM
Continual (S3)	Adam, lr 10^{-4} , 5 epochs/task; replay 15/class; LwF weight 3.0, $T=5$; EWC $\lambda=2800$
Distillation (S4)	Adam, lr 10^{-4} , 15 epochs; $T=2$; CE/transfer 0.75/0.25; S4.4: proto 0.5, 3 KMeans prototypes/class; reduced set: 32 clips/class
Meta-train (S5)	5-way 3-shot, 60 episodes $\times$ 8 epochs; 40 sports; inner SGD 5 steps @ 0.05; meta-lr 10^{-4} ; last 4 blocks trained
Adaptation (S5)	8 clips/class, 5 epochs, lr 3×10^{-4} ; replay 8/class; LwF 0.5, $T=2$
CLIP (S6)	ViT-B/32 frozen; $k \in \{1, 2, 4, 8, 16\}$; hybrid 0.4/0.6; probe $C=0.01$; MLP AdamW 10^{-3} , 50 epochs; baseline Adam 10^{-4} , 10 epochs
Capstone	replay 15/class (10 $\rightarrow$ 20 $\rightarrow$ 30); cosine KD, 10 epochs; MAML 5-way 5-shot, 35 sports (1 clip/video); YouTube: 10 clips/sport, 1/2/3/5 epochs, lr 3×10^{-4} , replay 8/class

parameters and, in the backbone stage, the inference time per clip.

IV. STAGE 1 — DATA SPLIT AND LEAKAGE

UCF101 natively collects clips into groups. All clips of a group share the same actor, background and camera angle. If one splits data per clip, segments of the *same* recording can land in train and test, and the model can easily recognise the *scene context* instead of the *action*. We therefore assign each whole group to a single split (panel S1 of Fig. 1).

Results and interpretation. As shown in Table II, the shipped clip-level split places 400 of the 500 source groups (80%) in multiple subsets, allowing closely related clips

TABLE II
SOURCE GROUPS SHARED BETWEEN SPLITS (20 STUDIED CLASSES, 2,679 VIDEOS).

Split	Groups	Groups in >1 split	Leak
Shipped (per clip)	500	400	80%
Group-disjoint	500	0	0%

TABLE III
BACKBONE BAKE-OFF ON THE 10 BASE CLASSES: TEST ACCURACY (%) ON UCF101’S SHIPPED SPLIT (171 TEST CLIPS, LEAKING) AND ON OUR GROUP-DISJOINT SPLIT (212 TEST CLIPS). SAME BUDGET FOR EVERY BACKBONE; MS/CLIP IS THE INFERENCE TIME PER CLIP. –: NOT MEASURED.

Backbone ϕ	Params (M)	ms/clip	Shipped	Group-disjoint
ScratchCNN	103.1	10.2	89.5	52.8
ResNet18	12.0	15.7	97.1	90.1
DenseNet121	8.3	53.1	98.8	92.0
VGG19-BN	20.8	115.7	96.5	92.0
ViT-B/16	87.4	–	99.4	–
ResNet50 (selected)	25.9	54.6	98.2	93.9

from the same recording to appear in both training and evaluation data. By contrast, the group-disjoint split eliminates this overlap while preserving the standard 72/12/16% train/validation/test proportions. Consequently, the accuracies reported in subsequent stages may be lower than results obtained with the shipped split, but they are not inflated by source-group leakage. Stage 2 measures the size of this inflation: the same backbones score 4.3 to 36.7 points higher on the shipped split (Table III). Applying this correction at the data partitioning stage therefore provides a consistent, leakage-free evaluation protocol for the entire study.

V. STAGE 2 — BACKBONE SELECTION

In this section we evaluate six different backbones in order to select the best one to perform action-recognition on UCF101. The tested backbones include: a simple CNN trained from scratch, four CNNs pre-trained on ImageNet (ResNet18, DenseNet121, VGG19-BN, and ResNet50), and a vision transformer, ViT-B/16, also pre-trained on ImageNet. Each backbone ϕ is injected in the following pipeline: video clip $\rightarrow \phi \rightarrow \text{LSTM} \rightarrow \text{head}$, as shown in panel S2 of Fig. 1; for the ViT, ϕ is the 768-d class token of every frame, followed by a two-layer LSTM. Each resulting architecture is fine-tuned over 10 UCF101 classes under the same training conditions (Adam, learning rate 10^{-4} , 5 epochs): the pretrained backbones keep their early layers frozen and fine-tune only their last block (the last two of the twelve encoder blocks for the ViT), together with the LSTM and the head; the custom CNN is trained entirely. The ViT was trained in an earlier version of the pipeline that used UCF101’s shipped split. To compare it fairly, Table III also reports every CNN, trained with the same budget, on that same split.

Results and selection rationale. On our group-disjoint split (Table III, last column), the from-scratch CNN has the

largest parameter count (103.1M) yet the lowest test accuracy (52.8%). Its size is primarily due to the LSTM receiving a high-dimensional flattened feature map, illustrating that model capacity alone cannot compensate for the absence of pretrained spatial features in this limited-data regime. In contrast, all four pretrained CNNs achieve more than 90% accuracy.

The shipped split inflates every result. On UCF101’s shipped split the same models score 4.3 to 36.7 points higher (ResNet50: 98.2% instead of 93.9%), and the ranking changes: DenseNet121 (98.8%) appears better than ResNet50. The inflation is largest for the from-scratch CNN (89.5% instead of 52.8%): without pretrained features it memorises the scene context that the leaking test clips share with the training clips. A leaking split can therefore hide a weak model and reorder the strong ones, as anticipated in Stage 1.

Vision transformer. On the shipped split the ViT-B/16 is the most accurate backbone (99.4%). However, it differs from ResNet50 by only 2 of the 171 test clips, it has $3.4\times$ more parameters (87.4M against 25.9M), and it has not yet been evaluated on the group-disjoint split, where its advantage would have to be measured again.

We select **ResNet50** because it obtains the highest accuracy on the leakage-free split (93.9%), provides a 2048-dimensional feature representation for the subsequent stages, and requires 54.6 ms per clip, approximately half the latency of VGG19-BN (115.7 ms per clip).

VI. STAGE 3 — CONTINUAL LEARNING

In this section we aim to extend the knowledge of the selected ResNet50+LSTM model, passing from 10 old classes to 20, increasing the count of 5 new classes every time. We tested five different methods, that approach the problem of protecting old knowledge in different ways, modifying the base cross-entropy loss of the task using unique strategies:

- **S3.0 Naive:** trains only on the new classes using $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{new})$, with no explicit mechanism for preserving knowledge of the old classes.
- **S3.1 EWC** [1]: adds the penalty $\frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$. Here, F_i estimates the importance of parameter θ_i to the previous model, and θ_i^* is its previous value. The penalty discourages large changes to important parameters.
- **S3.2 LwF** [3]: adds the distillation loss $\lambda \text{KD}(z_{1:C_{\text{old}}}, z_{1:C_{\text{old}}}^*)$. On the new-class clips, this loss encourages the current model’s predictions for the old classes to remain close to those produced by a frozen copy of the previous model.
- **S3.3 Replay:** stores 15 old clips per class and forms each training batch with equal numbers of new and replayed clips. The model is trained with $\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{new} \cup \text{old})$.
- **S3.4 Replay+LwF** [4]: combines replayed old-class clips with the LwF distillation loss.

Table IV reports the final accuracy of each method, while Table V shows how performance changes after each task.

Results and interpretation. *S3.0 Naive* learns each new task well (98.2% on task 1 and 94.9% on task 2), but the base-class accuracy falls from 93.9% to 22.6% after the first update

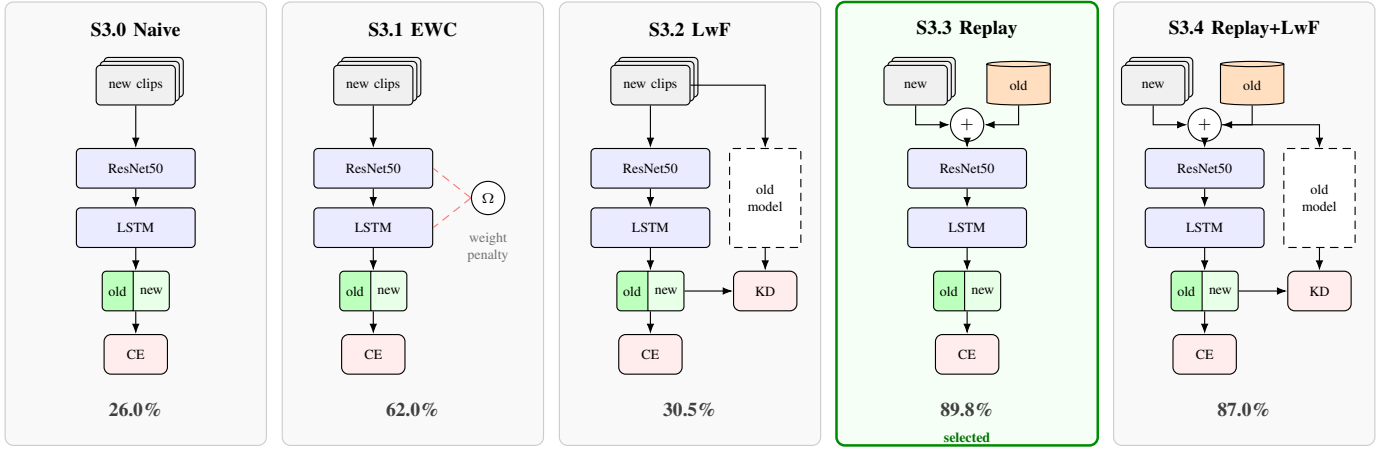


Fig. 3. **Stage 3: the five continual-learning methods**, top to bottom from the input clips to the loss. All train the same network (blue) with a head that grows by the new classes; they differ only in what protects the old classes: nothing (S3.0), a weight penalty $\Omega = \frac{\lambda}{2} \sum_i F_i(\theta_i - \theta_i^*)^2$ (S3.1), the scores of a frozen copy of the old model (S3.2, dashed), stored old clips (S3.3, orange), or both (S3.4). Bottom: final accuracy on all 20 classes (Table IV).

TABLE IV
CONTINUAL LEARNING AFTER THE FULL $10 \rightarrow 15 \rightarrow 20$ STREAM (%).
FINAL: ALL 20 CLASSES; OLD: CLASSES 0–14; NEW: LAST TASK 15–19;
BASE: ORIGINAL 10.

Method	Final	Old (0–14)	New (15–19)	Base (0–9)
S3.0 Naive	26.0	0.9	94.9	0.5
S3.2 LwF	30.5	5.3	100.0	0.0
S3.1 EWC	62.0	71.1	36.8	76.9
S3.4 Replay+LwF	87.0	82.6	99.2	83.0
S3.3 Replay	89.8	86.7	98.3	89.2

TABLE V
WHEN EACH METHOD FORGETS: ACCURACY PER TASK (%) AFTER
LEARNING TASK 1 AND AFTER LEARNING TASK 2. BEFORE ANY NEW
TASK THE BASE ACCURACY IS 93.9.

Method	after task 1		after task 2		
	Base	Task 1	Base	Task 1	Task 2
S3.0 Naive	22.6	98.2	0.5	1.8	94.9
S3.1 EWC	80.7	79.1	76.9	60.0	36.8
S3.2 LwF	26.4	97.3	0.0	15.5	100.0
S3.3 Replay	89.6	94.6	89.2	81.8	98.3
S3.4 Replay+LwF	87.3	96.4	83.0	81.8	99.2

and to 0.5% after the second. Because training batches contain only new-class clips, the shared parameters are updated without any signal that preserves the old decision boundaries. S3.2 LwF shows a similar pattern: its base accuracy decreases from 26.4% after task 1 to 0.0% after task 2. In the original LwF setting [3], tasks use separate output heads and the task identity is known at inference. Our setting is more demanding because a single growing head must distinguish all classes without knowing the task. Moreover, the distillation loss is computed only over the old-class outputs on new clips. It preserves their relative responses but does not directly train the model on old examples or calibrate old classes against the new ones. Consistent with this limitation, the confusion matrices show

that most forgotten base clips are assigned to new classes. S3.1 EWC provides greater stability, retaining 76.9% accuracy on the base classes, but its task 2 accuracy reaches only 36.8%. The parameter penalty therefore limits forgetting at the cost of reduced plasticity for learning the newest classes. S3.3 Replay achieves the best final accuracy (89.8%) and maintains strong performance across the base classes (89.2%), task 1 (81.8%), and task 2 (98.3%). The stored clips provide direct supervision for old classes and keep them represented in each training batch. S3.4 Replay+LwF also performs well, but its final accuracy is slightly lower (87.0%). In this experiment, adding distillation to replay provides no additional benefit and reduces old-class retention relative to replay alone. **Choice for the pipeline: replay** (panel S3 of Fig. 1). For this video stream, replaying a small number of stored clips is more effective than the tested data-free regularization methods.

VII. STAGE 4 — KNOWLEDGE DISTILLATION

In this section we aim to condense the 25.9M-parameter teacher into a smaller device, to be deployed even under computational and architectural constraints. We chose as student MobileNetV3-Small+LSTM (≈ 1.8 M parameters, $14\times$ smaller). The five objectives in Fig. 4 use the same ImageNet-pretrained student initialization, training data, computational budget, and random seed. Except for proto-alignment, they use $\mathcal{L} = 0.75 \mathcal{L}_{\text{CE}} + 0.25 \mathcal{L}_{\text{transfer}}$:

- **S4.0 CE only**: sets $\mathcal{L}_{\text{transfer}} = 0$ and trains without a teacher, providing the baseline.
- **S4.1 KD** [6]: matches the softened output distributions of the teacher and student using $T^2 \text{KL}(\sigma(z_t/T) \parallel \sigma(z_s/T))$, with $T = 2$.
- **S4.2 Cosine**: aligns the directions of the 256-dimensional clip features, $1 - \cos(h_s, h_t)$.
- **S4.3 MSE** [7]: uses a learnable regressor g to map the student feature h_s into the teacher’s feature space, then minimizes $\|g(h_s) - h_t\|_2^2$. The regressor is removed at inference.

TABLE VI

DISTILLATION INTO THE MOBILENETV3-SMALL STUDENT (20-CLASS GROUP-DISJOINT TEST, %). SAME PRETRAINED STUDENT, BUDGET, SEED AND INITIALISATION; ONLY THE TRANSFER TERM DIFFERS. FULL: ALL 1,917 TRAINING CLIPS (≈ 96 PER CLASS); 32/CLASS: 640 TRAINING CLIPS.

Model	Full	32/class	Params (M)
Teacher (ResNet50, Replay+LwF)	87.0	86.8	25.9
S4.0 Student, CE only	80.6	61.5	1.8
S4.1 Student, CE + KD	79.7	62.0	1.8
S4.2 Student, CE + cosine	82.7	59.7	1.8
S4.3 Student, CE + MSE (g)	75.9	61.3	1.8
S4.4 Student, proto-align (new)	80.4	67.9	1.8

- **S4.4 Proto-align** (proposed): before training, the teacher features for each class are grouped by k -means into three *prototypes*. During training, each student feature is pulled toward the nearest teacher prototype for its class using $\min_j \|h_s - \mu_{y,j}\|_2^2$. This term is added to CE and KD: $\mathcal{L} = 0.75 \mathcal{L}_{\text{CE}} + 0.25 \text{KD} + 0.5 \mathcal{L}_{\text{proto}}$. Unlike clip-level feature matching, this objective transfers a class-level summary of the teacher’s representation, following the general idea of prototypical networks [13].

Because the pretrained student can already learn strong features from class labels, we also repeat the study with a *reduced training set* of 32 clips per class (640 clips, approximately one third of the full 1,917), while keeping the test set unchanged. The teacher is the same Stage-3 model trained on the full dataset; it therefore has access to more training examples than the student, a setting in which knowledge transfer is expected to be more useful. Fig. 5 illustrates proto-alignment in detail.

Results and interpretation. Table VI shows that the best full-data student is approximately $14\times$ smaller than the teacher and remains within 4.3 percentage points of its accuracy (82.7% versus 87.0%). The CE-only baseline is already strong (80.6%), partly because the student uses an ImageNet-pretrained backbone. S4.1 KD reaches 79.7% and therefore provides no improvement over CE alone in this setting. S4.2 Cosine performs best, reaching 82.7%, or 2.1 points above the baseline. This objective aligns the directions of the teacher and student clip features without requiring their magnitudes to match. S4.3 MSE is the weakest full-data method (75.9%). Matching both the direction and magnitude of features produced by different backbones may impose a more restrictive target on the student. Because these results use one seed and 439 test clips, small differences should be interpreted with caution. The main result of this stage is therefore the compression itself, with cosine distillation providing the best measured transfer objective. **Choice for the pipeline: cosine distillation** (panel S4 of Fig. 1).

Reduced-data setting. When training is limited to 32 clips per class, all full-data methods lose substantial accuracy; for example, the CE baseline decreases from 80.6% to 61.5%. Training accuracy still reaches 97–99%, while test accuracy remains near 60%, indicating overfitting. KD (62.0%) and MSE (61.3%) perform similarly to CE, whereas cosine reaches

59.7%. These objectives transfer a separate teacher target for each available training clip and provide little improvement when the student sees only a small set of clips. In contrast, S4.4 Proto-align achieves the highest accuracy at 67.9%: 6.4 points above CE and 5.9 points above KD. It also reduces the gap to the teacher from 25.3 to 18.9 points. This gain is larger than the approximate binomial standard error of 2.3 points on 439 test clips, although experiments with additional seeds are needed to measure training variability.

Proto-alignment differs from the other objectives because it transfers a class-level summary built from the teacher’s full set of features. Pulling each student feature toward the nearest of three class prototypes encourages samples from the same class to form compact groups, while multiple prototypes preserve variation in how an action can be performed. These results suggest that, when training data are scarce, class-level teacher targets are more useful than clip-level targets. With the full training set, however, proto-alignment reaches 80.4%, slightly below CE alone (80.6%) and 2.3 points below cosine (82.7%); when every class has about 96 training clips, the class-level targets add little that the labels do not already provide. Cosine therefore remains the objective of the full-data pipeline (panel S4 of Fig. 1).

VIII. STAGE 5 — FEW-SHOT META-ADAPTATION

Design. Given the 20-class student, how well can it learn *five brand-new sports* from only a few clips? Every arm uses the same adaptation—the same 8 clips per new sport, 5 epochs, learning rate and seed, a head grown from 20 to 25 classes—so only two things vary (Fig. 6): the *starting point* and the *protection* of the old classes. The three starting points are:

- **S5.0 Plain:** the distilled student as it is.
- **S5.1 Fine-tune:** the student after ordinary supervised training on 40 other sports. This is the control: it sees the same extra data as MAML.
- **S5.2 MAML** [8]: the student after first-order MAML on the same 40 sports. Each episode draws 5 sports with 3 support and 3 query clips; the *inner* loop adapts a copy on the support clips, $\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}^{\text{sup}}(\theta)$ (5 SGD steps), and the *outer* loop moves the start by the loss of the adapted copy on the query clips, $\theta \leftarrow \theta - \beta \nabla_{\theta'} \mathcal{L}^{\text{qry}}(\theta')$.

For S5.1 and S5.2 only the last four backbone blocks, the LSTM and the head are trained, and a fresh 20-class head is then re-fitted to the new features (a short linear probe on 12 clips per class). Each start is adapted with no protection, with replay (8 old clips per class), or with replay+LwF, giving nine arms ranked by the harmonic mean of Eq. (2).

Results. Table VII ranks the nine arms; Table VIII shows how the new-sport accuracy grows with the number of clips. Before adaptation the student scores 75.9% on the base classes; during meta-training the query accuracy rises from 0.83 to 0.98 over 8 epochs, so the start is really being reshaped.

Why each arm behaves as it does. Without protection every start learns the new sports but wipes the base (base 0.011–0.189): with only new clips, the whole network is rewritten, exactly as in naive continual learning. *MAML alone*

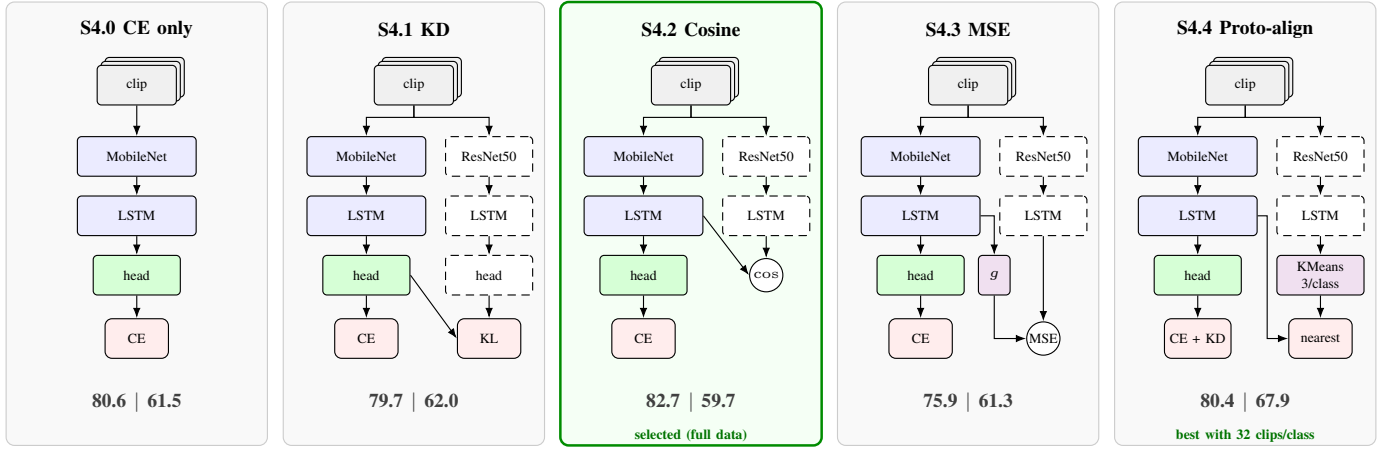


Fig. 4. **Stage 4: the five distillation objectives**, top to bottom. The pretrained student (blue, left of each panel) always learns from the labels (CE); the frozen teacher (dashed, right) adds nothing (S4.0), its softened scores (S4.1), its clip feature by direction (S4.2), its clip feature through a learnable regressor g (S4.3), or KD plus the nearest of 3 KMeans prototypes of the student’s class (S4.4, new; detail in Fig. 5). Bottom: student test accuracy (%) with the full training set | with 32 clips per class (Table VI).

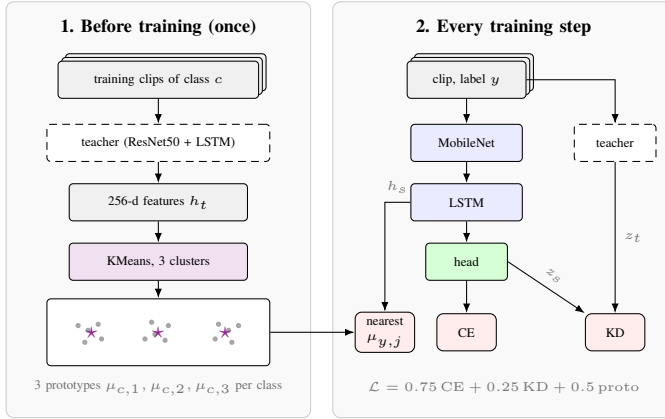


Fig. 5. **S4.4 Proto-align in detail**. (1) Once, before training, the frozen teacher encodes the training clips of every class and KMeans groups their 256-d features into 3 prototypes per class (stars). (2) At every step the student learns from the label (CE), from the teacher’s softened scores (KD), and from a prototype term that pulls its clip feature h_s to the nearest prototype of its own class y , $\min_j \|h_s - \mu_{y,j}\|^2$. The teacher and the prototypes are never updated; only the student (blue) trains.

learns the new sports best (0.661), because its start is built to move quickly onto a new task—and with nothing holding it back it moves all the way. *Replay* fixes the base in every case (0.576 to 0.724) but slows new learning, most for the plain start (0.295). *MAML+replay* combines a start that learns fast with stored clips that hold the old classes, and is the best balance (H 0.600). The key comparison is with the control: *fine-tune+replay* saw the same 40 sports and reaches only 0.530, so the extra 0.07 comes from meta-learning, not from the extra data. The LwF variants are always slightly worse: with only a few new clips, keeping the old scores fixed also holds back the fresh new-class rows. Table VIII shows that MAML’s advantage is a property of the *start*: it is largest with 1 clip (0.26 against 0.00–0.03) and disappears by 10 clips (0.65 against 0.66 for the plain start), when there are enough

TABLE VII
FEW-SHOT META-ADAPTATION: 20 BASE + 5 NEW SPORTS, 8 CLIPS PER NEW SPORT, IDENTICAL ADAPTATION FOR EVERY ARM, RANKED BY THE HARMONIC MEAN H (EQ. 2).

Start	Protection	New	Base	H
S5.2 MAML	replay	0.625	0.576	0.600
S5.2 MAML	replay+LwF	0.580	0.542	0.561
S5.1 Fine-tune	replay	0.464	0.617	0.530
S5.1 Fine-tune	replay+LwF	0.446	0.626	0.521
S5.0 Plain	replay	0.295	0.677	0.411
S5.0 Plain	replay+LwF	0.223	0.724	0.341
S5.0 Plain	none	0.527	0.189	0.278
S5.2 MAML	none	0.661	0.011	0.022
S5.1 Fine-tune	none	0.500	0.011	0.022

TABLE VIII
NEW-SPORT ACCURACY VERSUS THE NUMBER OF CLIPS PER NEW SPORT.

Start	Protection	1	3	5	10
S5.0 Plain	none	0.00	0.13	0.43	0.66
S5.1 Fine-tune	none	0.03	0.37	0.54	0.59
S5.2 MAML	none	0.26	0.42	0.55	0.65
S5.0 Plain	replay	0.00	0.04	0.23	0.45
S5.1 Fine-tune	replay	0.00	0.22	0.34	0.49
S5.2 MAML	replay	0.16	0.19	0.44	0.55

steps for any start to learn. **Choice for the pipeline: MAML meta-training, adapted with replay** (panel S5 of Fig. 1).

IX. STAGE 6 — FEW-SHOT LEARNING WITH A FROZEN CLIP ENCODER

Through Stage 5 we developed a strategy to make our model able to adapt quickly to new tasks. We evaluated also a different approach, borrowing a *foundation model* pre-trained over a vast amount of data and keeping it frozen, using it as a 0-shot feature extractor. We chose CLIP [12] (ViT-B/32),

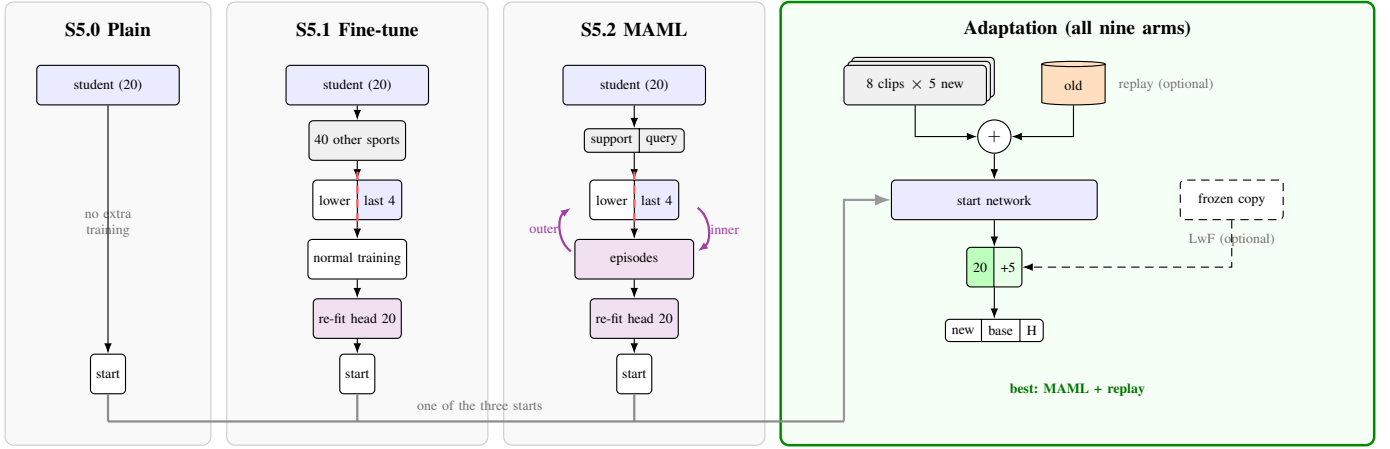


Fig. 6. **Stage 5: three starting points and one shared adaptation**, top to bottom. The plain student (S5.0), the student after normal training on 40 other sports (S5.1, the control), and the student after MAML on the same sports (S5.2, violet: inner/outer loop) are each adapted in the same way: 8 clips of 5 new sports, a head grown from 20 to 25 classes, with optional replay (orange) and optional LwF (dashed). Red dashed: only the last 4 blocks, LSTM and head are trained before adaptation.

a vision-language model already trained on 400M image-text pairs. Our goal is to test whether it can learn our actions from a handful of clips, or even from their names alone. This study is a reference for Stage 5; it is not part of the pipeline of Fig. 1.

Architectural design (Fig. 7). For every video clip, all 16 frames are passed through the frozen CLIP image encoder, to extract 16 embeddings. These are averaged to obtain one normalised 512-d clip feature map f . The task is to learn 10 base classes, across different few-shot scenarios: $k \in \{1, 2, 4, 8, 16\}$ training clips per class. The model is tested on the 212 clips of the group-disjoint test set. We compare four different methods:

- **S6.0 Baseline**: our ResNet50+LSTM model. The ImageNet backbone is frozen, while the LSTM and classifier head are trained on the k clips per class (Adam, 10^{-4} , 10 epochs).
- **S6.1 Hybrid prototypes**: the prompt “a video of a person performing $\langle \text{class} \rangle$ ” is encoded by the frozen CLIP text encoder to obtain a text prototype t_c . It is combined with the mean CLIP feature m_c of the k training clips for class c to form the normalised prototype $p_c = 0.4 t_c + 0.6 m_c$. Each test clip is assigned to the class with the highest cosine similarity $\cos(f, p_c)$. No parameters are trained; however, because the method uses the k labelled clips, it is a few-shot method rather than a purely zero-shot classifier.
- **S6.2 Linear probe**: a logistic-regression classifier is trained on the frozen clip features f from the k clips per class, using $C = 0.01$.
- **S6.3 MLP adapter**: a small MLP is trained on the frozen clip features f . It contains a $512 \rightarrow 256$ linear layer, batch normalisation, ReLU, dropout of 0.3, and a classification layer (AdamW, 10^{-3} , weight decay 10^{-2} , 50 epochs).

S6.0 and S6.3 are evaluated for every value of k , while S6.1 and S6.2 are evaluated only for $k = 16$.

Results and interpretation. Table IX shows that frozen

TABLE IX
STAGE 6: TEST ACCURACY (%) ON THE 10 BASE CLASSES (212 TEST CLIPS, CHANCE 10%) WITH k TRAINING CLIPS PER CLASS. “TRAINED” COUNTS THE TRAINABLE PARAMETERS; THE CLIP IMAGE ENCODER ($\approx 88\text{M}$) IS FROZEN. —: NOT RUN.

Method	Trained	$k=1$	2	4	8	16
S6.0 ResNet50+LSTM	2.4M	17.9	15.1	36.3	50.9	61.3
S6.1 Hybrid proto.	0	—	—	—	—	90.1
S6.2 Linear probe	5k	—	—	—	—	89.6
S6.3 MLP adapter	0.13M	75.0	73.1	74.1	79.2	86.8

CLIP features are highly effective in the few-shot setting. With one training clip per class, the MLP adapter reaches 75.0% accuracy, compared with 17.9% for ResNet50+LSTM. The comparison is not fully controlled because the models are trained for different numbers of epochs, but the large gap indicates the value of CLIP’s pretrained representation.

Increasing k from 1 to 4 provides little benefit to the MLP adapter (73.1–75.0%) because these clips come from the same recording and contain limited additional variation. Its accuracy rises to 79.2% with 8 clips and 86.8% with 16 clips, when examples from more recordings are included. This suggests that recording diversity is more useful than adding highly similar clips.

At $k = 16$, the simpler classifiers perform best: hybrid prototypes reach 90.1% and the linear probe reaches 89.6%, compared with 86.8% for the MLP adapter. Both are also close to the 93.9% accuracy of the Stage-2 model trained on the full dataset. These results show that strong frozen features require only a lightweight classifier; with 160 training clips, the additional capacity of the MLP does not improve generalisation.

Comparison between MAML (Stage 5) and CLIP (Stage 6). Both approaches address the same problem learning a new class from a few clips but in opposite ways, and each is

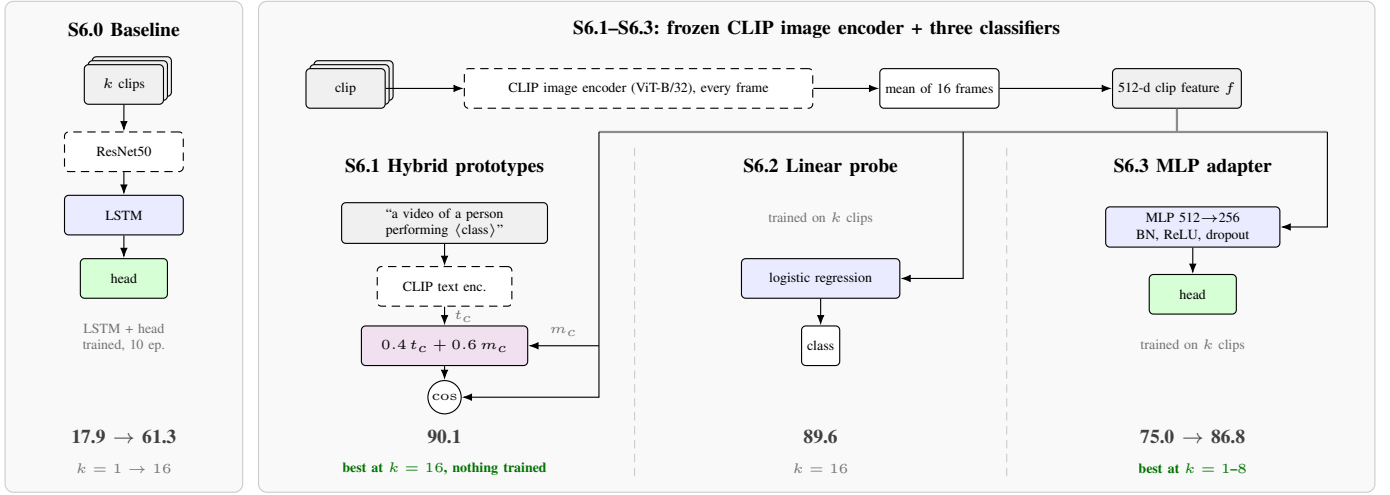


Fig. 7. **Stage 6: few-shot learning with a frozen CLIP encoder**, top to bottom. Left: our ResNet50+LSTM, its LSTM and head trained on the same k clips. Right: the frozen CLIP image encoder (dashed) turns every clip into one 512-d feature f (average of its 16 frames), and three classifiers use it: text + visual prototypes (S6.1; t_c from the class name, m_c the mean feature of the k clips of class c), a linear probe (S6.2), and a small MLP adapter (S6.3). Blue: trained on the k clips. Bottom: test accuracy (%), from 1 to 16 clips per class where the whole sweep was run.

preferable under different criteria:

- *Prior knowledge.* CLIP relies on representations learned from 400M web image-text pairs, which explains why a single clip per class already yields 75.0% accuracy. MAML, in contrast, only knows the UCF101 sports on which it was meta-trained, and still has to update the network to learn a new class.
- *Forgetting.* With frozen CLIP features, adding a class only requires adding a prototype or a row of a small classifier; the old classes are left untouched, so nothing is forgotten and no replay buffer is required. MAML updates the network and therefore needs replay to preserve the old classes.
- *Model size.* The frozen CLIP image encoder has approximately 88M parameters, about $50\times$ more than the 1.8M-parameter distilled student and more than the 25.9M-parameter teacher. It therefore gives up the compression obtained in Stage 4.
- *Temporal modelling.* Averaging frame features discards their order, whereas our recogniser uses an LSTM. The 10 base classes can largely be distinguished from the scene and objects of a single frame (e.g., a horse, a pommel horse, or a tabla), which favours CLIP; actions that differ mainly in their motion would test this limitation.

The two studies are not evaluated on the same task: Stage 5 adds 5 new sports to 20 known classes and is scored with the harmonic mean, whereas Stage 6 learns 10 classes from scratch. Their results therefore cannot be compared directly. A like-for-like comparison would require evaluating the CLIP classifiers (S6.1 and S6.3) on the YouTube test of Section X, next to the MAML student; we leave this evaluation to future work.

X. THE ULAL PIPELINE ON REAL VIDEO

Final pipeline In this section we aim to assemble the final pipeline, selecting the best performing strategy for each stage (backbone selection, continual learning, knowledge distillation, few-shot meta learning adaptation). Fig. 1 illustrates such pipeline, which executes the following protocol, over a $10 \rightarrow 20 \rightarrow 30$ classes stream:

- 1) **S1 Data preparation:** all video clips are pre-processed and splitted into train/val/test avoiding data leakage.
- 2) **S2 Teacher:** the ResNet50+LSTM teacher is trained on 10 base classes.
- 3) **S3 CL with replay strategy:** the teacher’s head grows to adapt sequentially to 20 and then 30 classes; every training batch mixes new clips with few stored old clips (15 per class) to prevent forgetting.
- 4) **S4 Cosine distillation:** the frozen 30-class teacher is distilled into the MobileNet student with CE + cosine strategy.
- 5) **S5 MAML:** the student is meta-trained with first-order MAML on 35 additional sports, i.e., the 40 meta-training sports of Stage 5 excluding the 5 that are already among the 30 known classes. Episodes are 5-way 5-shot, and only the last four backbone blocks, the LSTM, and the head are trained. Each meta-training clip is taken from a different source video, so the query clips always come from videos other than those of the support clips. The 30-class head is then re-fitted on 12 clips per class.
- 6) **Real-video test:** we collect YouTube videos of five sports that are unseen by all models: golf swing, front crawl, hammer throw, handstand push-ups, and handstand walking. For each sport, we use two videos for adaptation and reserve two separate videos for testing, with each video divided into 10 clips. One front-crawl video was unavailable, so that class is learned from a

TABLE X
THE PIPELINE ON UCF101 (GROUP-DISJOINT TEST).

Stage	Model	Classes	Acc. (%)	Params
S2	ResNet50 teacher	10	93.9	25.9M
S3	teacher + replay	30	79.7	25.9M
	old / new classes		78.8 / 81.4	
S4	student, cosine KD	30	67.2	1.8M
S5	MAML student (re-fitted)	30	60.8	1.8M

TABLE XI
LEARNING 5 UNSEEN SPORTS FROM 10 YOUTUBE CLIPS PER SPORT (WITH REPLAY), BY NUMBER OF PASSES OVER THE CLIPS (UPDATES IN BRACKETS). NEW: ACCURACY ON THE 100 CLIPS OF THE 10 UNSEEN YOUTUBE TEST VIDEOS; OLD: ACCURACY KEPT ON THE 30 UCF CLASSES (BEFORE ADAPTATION: TEACHER 0.797, KD STUDENT 0.672, MAML STUDENT 0.608); H: HARMONIC MEAN. BOLD: THE BETTER OF THE TWO STUDENTS (SAME SIZE, 1.8M PARAMETERS) IN EACH ROW; THE TEACHER (25.9M) IS SHOWN AS A REFERENCE.

Epochs	KD student			MAML student			Teacher (ref.)		
	New	Old	H	New	Old	H	New	Old	H
1 (13)	0.00	0.583	0.000	0.16	0.545	0.247	0.21	0.768	0.330
2 (26)	0.23	0.644	0.339	0.36	0.544	0.433	0.37	0.773	0.500
3 (39)	0.42	0.614	0.499	0.47	0.533	0.499	0.49	0.744	0.591
5 (65)	0.57	0.586	0.578	0.47	0.542	0.503	0.35	0.776	0.482

single source video. Each model expands its classifier from 30 to 35 classes and adapts on the same 10 clips per new sport, together with a replay buffer containing eight old clips per class. Adaptation uses 1, 2, 3, or 5 passes over the training clips, corresponding to 13, 26, 39, or 65 updates. The smaller budgets test the rapid adaptation for which MAML is designed. We compare the MAML student with the teacher and the plain distilled student under the same data, seed, and update budget. We report accuracy on the unseen YouTube test videos (new classes), retained accuracy on the 30 UCF classes (old classes), and the harmonic mean of the two.

Results on UCF101. Table X shows that replay remains effective on the longer stream. After two increments of 10 classes, the teacher reaches 79.7% accuracy over all 30 classes, with similar performance on the old (78.8%) and new classes (81.4%). The distilled student reaches 67.2% with approximately $14\times$ fewer parameters. This gap is larger than in Stage 4, although the experiments are not directly comparable because the stream contains more classes and uses fewer training epochs. After meta-training and re-fitting the classifier on 12 clips per class, the MAML student reaches 60.8%. Its 6.4-point reduction relative to the distilled student reflects the trade-off between retaining performance on known classes and preparing the model for rapid adaptation.

Results on real video. Table XI shows that the MAML student adapts faster than the plain distilled student under small update budgets. After one pass (13 updates), their new-class accuracies are 0.16 and 0.00, respectively. After two passes (26 updates), the MAML student reaches 0.36,

compared with 0.23 for the plain student and 0.37 for the much larger teacher. This result is consistent with the purpose of MAML: providing an initialization that can adapt in a small number of updates.

With more updates, the advantage decreases. After three passes, all models obtain new-class accuracies between 0.42 and 0.49. After five passes, the plain student performs best on both the new sports (0.57) and the harmonic mean ($H = 0.578$), while the MAML student remains at 0.47 on the new sports. This pattern is consistent with Stage 5 (Table VIII): meta-learning improves adaptation speed, but does not necessarily produce the highest accuracy after longer training.

Retention of old classes. Replay limits forgetting for all three models. Across the four update budgets, old-class accuracy ranges from 0.744 to 0.776 for the teacher, 0.583 to 0.644 for the distilled student, and 0.533 to 0.545 for the MAML student. The MAML student remains lower because its initial 30-class accuracy is also lower (0.608; Table X). Nevertheless, its harmonic mean exceeds that of the plain student after one and two passes, is equal after three passes, and is lower after five.

Teacher. The teacher retains the highest old-class accuracy and achieves the best harmonic mean through three passes ($H = 0.591$). Its new-class accuracy then decreases from 0.49 to 0.35 after five passes, while its old-class accuracy remains stable. This may indicate overfitting to the small adaptation set, but the limited test size prevents a firm conclusion.

These results have three important limitations. First, the real-video test contains only 10 videos (100 clips), so a single video can change new-class accuracy by as much as 0.10. Second, one front-crawl video was unavailable, leaving only one adaptation video for that class. Third, some classes exhibit a domain shift between adaptation and test videos; for example, the handstand push-up videos change from a studio to a gym setting.

XI. DISCUSSION AND LIMITATIONS

Four themes recur across the pipeline. First, *the evaluation protocol matters*: without the group-disjoint split, every accuracy would be inflated by scene memorisation (by 4.3 to 36.7 points in Stage 2), and without the harmonic mean the few-shot ranking would reward models that learn nothing new. Second, *stored data is the most effective protection against forgetting*: on video, rehearsal of a few clips per class outperforms every data-free method (EWC, LwF). LwF in particular was designed for a setting in which the task is known at test time, and it collapses when a single head must choose among all classes. Third, *each technique is useful in a specific regime*: the main benefit of distillation is compression, and with scarce data the teacher helps most through class-level prototypes rather than per-clip targets; meta-learning is most useful when new classes must be learned from very few clips and in very few updates, while with more clips or more updates the plain student catches up, both in Stage 5 and on real video. Fourth, *prior knowledge is the strongest few-shot tool, but it comes at the cost of model size*: frozen CLIP features learn the

base classes from a single clip (75.0%), where our recogniser needs many more, but the CLIP encoder is $50\times$ larger than the student and therefore targets a different deployment scenario, without size constraints, than the one considered by ULAL.

Limitations. All results are obtained with a single seed and on a single dataset, with short class streams (up to 30 classes). Rehearsal stores real clips, so its memory cost grows with the number of classes. The vision transformer has only been evaluated on the shipped split. The real-video test is small (5 sports, 10 test videos) and depends on downloadable YouTube videos (one of the 20 could not be downloaded); a larger set of videos and several seeds would be required to make the differences observed after three and five passes conclusive.

XII. CONCLUSION

In this work we presented ULAL, a compact real-video pipeline for lifelong action recognition that runs on a single free GPU. A leakage-free, group-disjoint split makes the reported results trustworthy, and a ResNet50+LSTM provides a strong teacher (93.9%). Among the continual-learning strategies, rehearsal is the most effective (89.8%, against 26.0% for naive fine-tuning). Cosine distillation produces a $14\times$ smaller student that retains 82.7% accuracy, while with scarce data the proposed prototype-alignment objective is the most effective way to use the teacher (+6.4 points over cross-entropy). Evaluated with a harmonic metric and a matched fine-tune control, MAML combined with replay is the best-balanced few-shot adapter ($H = 0.600$). As a reference, a frozen CLIP encoder learns the base classes from a single clip (75.0%, and 90.1% with 16 clips and no training), making it the best choice when model size is not a constraint. When the selected methods are combined into a single pipeline and tested on unseen YouTube videos, the meta-trained student is the faster of the two students: after 26 updates it matches the $14\times$ larger teacher on the new sports (0.36 against 0.37) and outperforms the plain student (0.23), which catches up only with more updates. Overall, each selected method contributes the capability for which it was chosen.

AUTHOR CONTRIBUTIONS

The work divides along the pipeline of Fig. 1, and Table XII indexes which author led which part. Each author owns a contiguous span of the pipeline together with a share of the shared infrastructure (the recogniser, the training harness, and the plotting and metrics code), and the two comparison-heavy stages (continual learning and meta-adaptation) sit with different authors. All three authors jointly designed the study, reviewed each other’s stages for fairness, and wrote the paper.

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TABLE XII
AUTHOR CONTRIBUTION INDEX, ORGANISED BY PIPELINE STAGE.

Contributor	Responsibilities
Second B. Author	S1 data split and leakage audit; S2 backbone bake-off; the shared recogniser and training harness.
Third C. Author	S3 continual learning (all five methods, forgetting timelines); S4 knowledge distillation (four objectives, compression analysis).
First A. Author	S5 few-shot meta-adaptation (MAML, fine-tune control, harmonic metric); the ULAL capstone and real-video test; the evaluation protocol and metrics.

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